

1) A software firm is developing an automated system to assign six developers to six different project modules AI, Web, Mobile, Cloud, Security, and Testing. Each developer must be assigned exactly one module, and no module can be assigned to more than one developer. However, due to expertise limitations and organizational policies, certain assignments are not allowed. For example, some developers are restricted from working on specific modules, while others can only be assigned to a limited subset. In addition, there are conditional dependencies between assignments for instance, assigning one developer to a module may enforce or restrict the assignment of another developer to a different module.

You are required to implement this system using a constraint-solving approach in Python. The solution should represent modules numerically, define decision variables for each developer, and apply all necessary constraints including uniqueness and conditional dependencies. The implementation must use a constraint solver to search for a valid solution and print the final assignment in a readable format. The program should also demonstrate how conditional constraints are enforced programmatically and ensure that no invalid combinations are included in the final output.

2) fitness analytics company has collected a dataset containing user activity information such as daily step count, workout duration, calories burned, heart rate, and sleep hours. The dataset contains missing values and potential inconsistencies due to sensor errors. The company wants to build a Python-based analysis pipeline that first cleans the dataset and then explores patterns in user behavior. After preprocessing, the system should generate visual insights such as distributions and relationships between variables. Furthermore, the company aims to group users into distinct categories based on their activity patterns without using any predefined labels.

You are required to write a program that loads the dataset, performs appropriate data cleaning, and conducts exploratory analysis using statistical summaries and visualizations. After that, apply a clustering technique to group similar users and visualize the resulting clusters. The program should also output the cluster centers and provide a brief interpretation of the clusters based on feature values.

3) A telecommunications company wants to predict whether a customer will leave its service based on historical data. The dataset includes features such as monthly charges, tenure, service type, and customer interaction metrics, along with a target variable indicating whether the customer churned. The dataset contains categorical features, missing values, and potential imbalance in the target classes.

You are required to develop a Python-based machine learning pipeline that preprocesses the dataset, converts categorical variables into numerical form, and splits the data into training and testing sets. After preprocessing, train at least two classification models and evaluate their performance on unseen data. The program should output accuracy metrics for both training and testing phases, compare model performance, and highlight which model is more suitable for this problem. Additionally, the implementation should consider how to handle imbalance or overfitting if observed during evaluation.